Replication of Climate Policy Dynamics Under Wealth Inequality

Risk, Homophily, and Evolutionary Dynamics

Elie Farah, Daniele Cioffi, Paolo Bordis

ULB

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Motivation

- Climate change mitigation is a collective-action problem
- Wealth inequality and asymmetric incentives complicate cooperation
- Key question:
  Does wealth inequality hinder or promote cooperation under risk?
- Vasconcelos et al. (2014) propose an evolutionary game-theoretic answer

Goals of This Project

- Independently replicate the main results of Vasconcelos et al. (2014)
- Implement the exact finite-population Markov process
- Reproduce:
  - Average group achievement (η_G)
  - Stationary distributions
  - Gradients of selection
  - Effects of obstinate agents

Model Overview

- Population of size Z
- Two wealth classes:
  - Rich: Z_R, endowment b_R
  - Poor: Z_P, endowment $b_P < b_R$
- Two strategies:

$$X \in \{C, D\}$$

- Random groups of size N

Collective-Risk Public Goods Game

- Cooperators contribute fraction c of endowment

$$c_R = cb_R, \quad c_P = cb_P$$

- Group succeeds if:

$$j_R c_R + j_P c_P \geq Mc\bar{b}$$

- If group fails:
  - Disaster occurs with probability r
  - All remaining endowment is lost

Payoffs

Defectors:

$$\Pi^{D}_{R/P} = b_{R/P}\left[\Theta(\Delta) + (1-r)(1-\Theta(\Delta))\right]$$

Cooperators:

$$\Pi^{C}_{R/P} = \Pi^{D}_{R/P} - c_{R/P}$$

- $\Theta(\cdot)$ is the Heaviside step function
- $\Delta = j_R c_R + j_P c_P - Mc\bar{b}$

Evolutionary Dynamics

- Finite population evolutionary process
- Pairwise comparison with Fermi rule:

$$P(X \to Y) = \frac{1}{1 + e^{\beta(f_X - f_Y)}}$$

- Homophily parameter h:
  - $h = 0$: anyone can imitate anyone
  - $h = 1$: imitation only within wealth class
- Mutation (exploration) with probability $\mu = 1/Z$

State Space and Markov Chain

- System state:

$$(i_R, i_P)$$

- i_R: number of rich cooperators
- i_P: number of poor cooperators
- Total states:

$$(Z_R + 1)(Z_P + 1)$$

- Exact transition matrix W constructed

Stationary Distribution

- Stationary distribution π satisfies:

$$W\pi = \pi$$

- Computed as eigenvector (eigenvalue $= 1$)

Aggregate Group Achievement

- For each state (i_R, i_P):

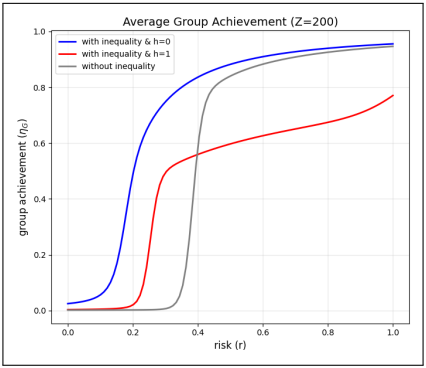
$$a_G(i_R, i_P) = P(\text{random group meets threshold})$$

- Average group achievement:

$$\eta_G = \sum_{i_R, i_P} \pi(i_R, i_P)\, a_G(i_R, i_P)$$

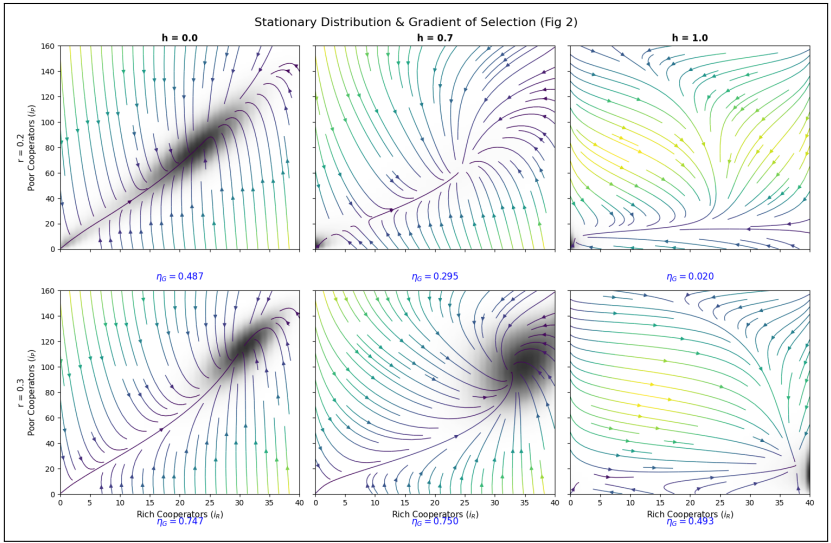
- Main welfare measure in the paper

Results I: Average Group Achievement



- Wealth inequality with $h = 0$ promotes early cooperation
- Homophily ($h = 1$) delays and weakens cooperation
- Sharp transition under wealth equality

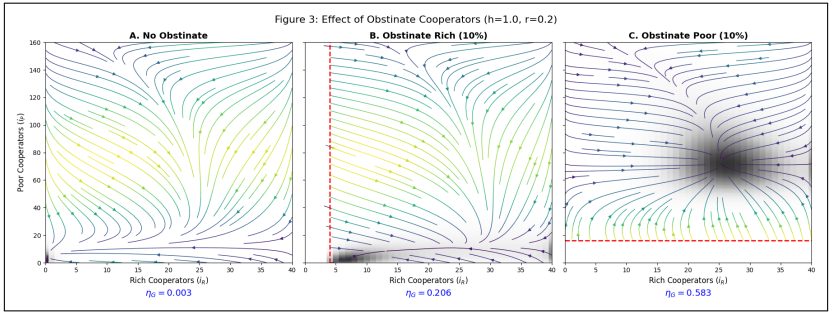
Results II: Stationary Distributions



Results II: Stationary Distributions

- Interior coexistence for $h = 0$
- Collapse of poor cooperation for $h = 1$, low r
- Risk restores cooperation among rich first

Results III: Obstinate Agents



- $h = 1$, $r = 0.2$
- Compare:
  - No obstinate agents
  - 10% obstinate rich
  - 10% obstinate poor

Main Takeaways

- Wealth inequality does *not* necessarily hinder cooperation
- Homophily is the primary destabilizing factor
- Poor agents play a disproportionate role
- Exact finite-population methods are powerful and revealing

Thank You

Questions?